

Optimal Transport as the Maximal Convex Extension of a Cost

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Abstract

The Kantorovich cost is often described as the convex extension of the pointwise cost from pairs of Dirac masses to pairs of probability measures. The precise statement contains an essential word: it is the *largest* weakly lower-semicontinuous jointly convex extension. Joint convexity, weak-star continuity and agreement on Dirac masses do not by themselves imply uniqueness. This note states the sharp probability-measure result, gives its short barycentric proof, exhibits a two-point counterexample to bare uniqueness, and records the measure-cone theorem of Savaré and Sodini.

1 The probability-measure statement

Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces and let $c \in C(\mathcal{X} \times \mathcal{Y})$ be finite. Endow $\mathcal{P}(\mathcal{X})$ and $\mathcal{P}(\mathcal{Y})$ with their narrow, equivalently weak-star, topologies. For $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, write $\Pi(\alpha, \beta)$ for the probability measures on $\mathcal{X} \times \mathcal{Y}$ with marginals α and β , and define

$$\mathcal{L}_c(\alpha, \beta) := \min_{\pi \in \Pi(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) \, d\pi(x, y). \quad (1)$$

Compactness and continuity ensure that the minimum is attained.

The geometry behind the result is elementary. Introduce the Dirac embedding

$$j : \mathcal{X} \times \mathcal{Y} \longrightarrow \mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{Y}), \quad j(x, y) = (\delta_x, \delta_y).$$

A probability measure π on $\mathcal{X} \times \mathcal{Y}$ belongs to $\Pi(\alpha, \beta)$ exactly when the barycenter of $j_{\#}\pi$ is (α, β) , in the weak sense

$$(\alpha, \beta) = \int_{\mathcal{X} \times \mathcal{Y}} (\delta_x, \delta_y) \, d\pi(x, y). \quad (2)$$

Thus couplings are precisely the convex decompositions of the pair of marginals into compatible pairs of extreme points.

Theorem 1.1 (Maximal convex-extension characterization). *The functional $\mathcal{L}_c$ has the following properties.*

1. *It is jointly convex and weak-star continuous on $\mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{Y})$.*
2. *It extends the pointwise cost:*

$$\mathcal{L}_c(\delta_x, \delta_y) = c(x, y) \quad \text{for every } (x, y) \in \mathcal{X} \times \mathcal{Y}.$$

3. *If $F : \mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{Y}) \rightarrow (-\infty, +\infty]$ is proper, jointly convex and weak-star lower semicontinuous, and*

$$F(\delta_x, \delta_y) \leq c(x, y) \quad \text{for every } (x, y), \quad (3)$$

then

$$F(\alpha, \beta) \leq \mathcal{L}_c(\alpha, \beta) \quad \text{for every } (\alpha, \beta). \quad (4)$$

Consequently, $\mathcal{L}_c$ is the pointwise largest jointly convex weak-star lower-semicontinuous functional dominated by c on Dirac pairs. Since $\mathcal{L}_c$ is continuous and has equality on Dirac pairs, it is also the largest member of the smaller class specified by the three properties in the question.

Proof. The only coupling of δ_x and δ_y is $\delta_{(x,y)}$, which proves the Dirac identity. If $\pi_i \in \Pi(\alpha_i, \beta_i)$ are optimal and $t \in [0, 1]$, then $t\pi_0 + (1-t)\pi_1$ couples $t\alpha_0 + (1-t)\alpha_1$ and $t\beta_0 + (1-t)\beta_1$. Its cost is the corresponding convex combination, proving joint convexity.

We briefly verify continuity. Suppose $(\alpha_n, \beta_n) \rightarrow (\alpha, \beta)$. Any convergent subsequence of optimal plans π_n has a limit in $\Pi(\alpha, \beta)$; continuity of c then gives

$$\mathcal{L}_c(\alpha, \beta) \leq \liminf_n \mathcal{L}_c(\alpha_n, \beta_n).$$

Conversely, start from an optimal $\pi \in \Pi(\alpha, \beta)$. On compact metric spaces, weak convergence is equivalent to convergence in the Wasserstein distance associated with any bounded compatible metric. Couple α_n to α and β_n to β with costs tending to zero, and glue these two couplings through π . This produces $\tilde{\pi}_n \in \Pi(\alpha_n, \beta_n)$ with $\tilde{\pi}_n \rightharpoonup \pi$. Uniform continuity of c gives

$$\limsup_n \mathcal{L}_c(\alpha_n, \beta_n) \leq \lim_n \int c d\tilde{\pi}_n = \int c d\pi = \mathcal{L}_c(\alpha, \beta).$$

It remains to prove maximality. Fix any $\pi \in \Pi(\alpha, \beta)$. Equations (2) and (3), together with Jensen's inequality, yield

$$F(\alpha, \beta) \leq \int F(\delta_x, \delta_y) d\pi(x, y) \leq \int c(x, y) d\pi(x, y).$$

For completeness, this version of Jensen follows by approximating π by finitely supported probability measures, applying finite-dimensional convexity, and passing to the limit using lower semicontinuity. Taking the infimum over π proves (4). $\square$

Equivalently, the theorem gives the convex-roof formula

$$\mathcal{L}_c(\alpha, \beta) = \inf \left\{ \int c(x, y) d\pi(x, y) : (\alpha, \beta) = \int (\delta_x, \delta_y) d\pi(x, y) \right\}. \quad (5)$$

The marginal constraints in optimal transport are therefore exactly the barycenter constraint in the convexification of the Dirac data.

2 Why the three properties do not imply uniqueness

The maximality clause cannot be omitted, even when all spaces are finite and “weak-star continuous” means ordinary continuity.

Example 2.1 (A two-point counterexample). Let $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and $c(x, y) = |x - y|$. Write

$$\alpha_p = (1-p)\delta_0 + p\delta_1, \quad \beta_q = (1-q)\delta_0 + q\delta_1.$$

Then

$$\mathcal{L}_c(\alpha_p, \beta_q) = |p - q|.$$

The alternative functional

$$F(\alpha_p, \beta_q) = (p - q)^2$$

is continuous and jointly convex because it is the square of a linear functional. For the four Dirac pairs $p, q \in \{0, 1\}$, it satisfies $F(\delta_p, \delta_q) = |p - q| = c(p, q)$. Nevertheless,

$$F(\alpha_{1/2}, \beta_0) = \frac{1}{4} < \frac{1}{2} = \mathcal{L}_c(\alpha_{1/2}, \beta_0).$$

Hence joint convexity, weak-star continuity and exact Dirac values imply only $F \leq \mathcal{L}_c$, not $F = \mathcal{L}_c$.

One may turn the maximality statement into a literal uniqueness axiom in either of two equivalent ways: require F to be pointwise maximal among the admissible convex extensions, or require F to equal the convex roof of its Dirac values. Either condition forces $F = \mathcal{L}_c$ by Theorem 1.1.

3 The Savaré–Sodini measure-cone theorem

The exact theorem of Savaré and Sodini is more general and cleaner from the viewpoint of convex analysis [1, Theorem 4.4]. Let $\mathcal{M}(\mathcal{X})$ denote the vector space of finite signed Radon measures, with weak topology induced by $C_b(\mathcal{X})$. Define the singular functional on $\mathcal{M}(\mathcal{X}) \times \mathcal{M}(\mathcal{Y})$ by

$$F_c(\mu, \nu) := \begin{cases} mc(x, y), & (\mu, \nu) = (m\delta_x, m\delta_y), \quad m \geq 0, \\ +\infty, & \text{otherwise.} \end{cases} \quad (6)$$

with the convention $0 \cdot (+\infty) = 0$. In the formula below, $\Pi(\mu, \nu)$ denotes the nonnegative finite measures on $\mathcal{X} \times \mathcal{Y}$ with marginals μ and ν . Extend the transport functional to the same vector space by

$$T_c(\mu, \nu) := \begin{cases} \min_{\gamma \in \Pi(\mu, \nu)} \int c \, d\gamma, & \mu, \nu \geq 0, \quad \mu(\mathcal{X}) = \nu(\mathcal{Y}), \\ +\infty, & \text{otherwise.} \end{cases} \quad (7)$$

Theorem 3.1 (Savaré–Sodini). *Let $\mathcal{X}$ and $\mathcal{Y}$ be completely regular spaces and let $c : \mathcal{X} \times \mathcal{Y} \rightarrow [0, +\infty]$ be lower semicontinuous and proper. Then T_c is the weakly lower-semicontinuous convex envelope of F_c . Equivalently, it is the largest weakly lower-semicontinuous convex functional G such that $G \leq F_c$. It is, in addition, sublinear and satisfies*

$$T_c(m\delta_x, m\delta_y) = mc(x, y).$$

On compact metric spaces, the proof is the same barycentric argument as above. If μ and ν have common mass $m > 0$ and $\gamma \in \Pi(\mu, \nu)$, then

$$(\mu, \nu) = \int_{\mathcal{X} \times \mathcal{Y}} (m\delta_x, m\delta_y) \, d\frac{\gamma}{m}(x, y).$$

Thus every convex $G \leq F_c$ satisfies

$$G(\mu, \nu) \leq \int G(m\delta_x, m\delta_y) \, d\frac{\gamma}{m} \leq \int c \, d\gamma,$$

and hence $G \leq T_c$. Convexity and lower semicontinuity of T_c give the reverse envelope inclusion. The general completely regular case requires the compactness theorem for plans with converging marginals established in [1].

There is also a one-line dual interpretation. The convex conjugate of F_c is the indicator of

$$\{(f, g) \in C_b(\mathcal{X}) \times C_b(\mathcal{Y}) : f(x) + g(y) \leq c(x, y)\}.$$

Fenchel–Moreau therefore identifies the biconjugate of F_c with the Kantorovich dual, while the envelope theorem identifies the same biconjugate with T_c . This is the convex-analytic core of Kantorovich duality in the Savaré–Sodini proof.

References

- [1] G. Savaré and G. E. Sodini. A simple relaxation approach to duality for optimal transport problems in completely regular spaces. *Journal of Convex Analysis*, 29(1):1–12, 2022. <https://cvgmt.sns.it/paper/5594/>.